Timer

學習目標

- 本章節介紹:
  1. 使用timer，使得經過一段時間後可以呼叫指定的函式

建立/取消一個timer

- 載入函式庫

  from threading import Timer
- 建立語法:

  t = Timer(秒數, 函式名稱, args=[])

  t.start()
- 或

  Timer(秒數, 函式名稱) .start()
- 取消語法:

  t.cancel ()

範例1 (函式無參數帶入)

```
from tkinter import *#載入函式庫
from threading import Timer

t = None
counter = 1
def change_time():
    global labelText
    global counter
    global t
    counter = counter + 1
    labelText.set(str(counter))
    t = Timer(1, change_time)
    t.start()

def pause():
    global root
    global t
    t.cancel()
    root.destroy ()

root = Tk()#建立視窗
labelText = StringVar()
labelText.set(str(counter))
Label(root, textvariable=labelText).pack ()
Button(root, text="Exit", command = pause).pack ()
t = Timer(1, change_time)
t.start()
root.mainloop()
```

範例1 (函式有參數帶入)

```
from tkinter import *#載入函式庫
from threading import Timer

t = None
counter = 1
def change_time(a, b):
    global labelText
    global counter
    global t

    counter = counter + 1
    labelText.set(str(counter))
    t = Timer(1, change_time, [a+1, b+1])
    print (a+1, " ", b+1)
    t.start()

def pause():
    global root
    global t
    t.cancel()
    root.destroy ()

root = Tk()#建立視窗
labelText = StringVar()
labelText.set(str(counter))
Label(root, textvariable=labelText).pack ()
Button(root, text="Exit", command = pause).pack ()
t = Timer(1, change_time, [1, 2])
t.start()
root.mainloop()
```